

A024 · Genetic Regulation of Circular RNAs Reveals a Distinct Molecular Layer Underlying Psychiatric Risk

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Circular RNAs (circRNAs) remain an underexplored layer of transcriptomic regulation in psychiatric disorders. We quantified circRNA expression from 1,022 [518 neurotypical, 365 schizophrenia (SCZ) and 139 bipolar disorder (BIP)] postmortem cortex samples from PsychENCODE consortium cohorts and integrated these profiles with matched linear RNA and genotype profiles. We identified 23 SCZ-associated and 3 BIP-associated differentially expressed circRNAs ($FDR < 0.05$; $FDR\text{-}circDEG$). We trained genetically regulated circRNA expression (circGReX) models using neurotypicals and applied them to SCZ and BIP GWAS to perform Transcriptomic Wide association analysis (TWAS) which identified 22 and 4 circGReX trait associations (circGTAs), respectively. Pathway enrichment of circDEGs and circGTAs implicated neuronal and synaptic processes for both disorders. In UK Biobank, circGReX-imaging associations were predominantly negatively correlated with SCZ and BIP circGTAs, but positively correlated with Alzheimer's disease circGTAs. circKLHL24 isoforms showed the most prominent imaging associations. Many co-expression modules containing our $FDR\text{-}circDEGs$ were enriched for psychiatric and neurodegenerative risk genes, including our identified circGTAs, and these modules were enriched for cognitive and neurodevelopmental traits. To conclude, circRNAs represent a distinct regulatory layer in psychiatric disorders, linking genetic risk to synaptic biology, brain structure and cognition through disease-specific expression, TWAS prioritization, and imaging associations.